

Ranga Reddy

Senior Data Platform & Lakehouse Engineer

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PROFESSIONAL SUMMARY

Senior Data Platform & Lakehouse Engineer with 14+ years of experience designing, optimizing, and scaling petabyte-scale distributed data platforms. Active open-source contributor to **Apache Hudi** and **Apache XTable**, specializing in indexing, compaction, clustering, and multi-writer concurrency models. Ex-Cloudera Backline Engineer serving as the final escalation point for global Fortune 500 P0/P1 Spark and Iceberg production outages. Deep expertise in Spark Catalyst query optimization, lakehouse table formats, real-time streaming, and cloud infrastructure across AWS and GCP.

TECHNICAL SKILLS

Lakehouse & Table Formats: Apache Hudi, Apache Iceberg, Hive Metastore, Apache XTable.

Distributed Compute & Query Engines: Apache Spark (SQL, DataFrames, Streaming), Trino, AWS Athena, Apache Hive, Apache Flink.

File Formats: Parquet, ORC, Avro, JSON.

Streaming & Messaging: Apache Kafka, Hudi Streamer (DeltaStreamer), Apache Storm, Spark Structured Streaming.

Data Engineering: ETL/ELT pipeline design, data ingestion, data modeling, CDC, partitioning and schema evolution, data quality, performance tuning, cost optimisation.

Cloud: AWS (S3, EMR, Glue, Athena), GCP (Dataproc, GCS, BigQuery), MinIO.

Platforms & Infrastructure: Cloudera CDP/CDH, Hortonworks HDP, Hadoop, HBase, Linux.

Languages: Java, Scala, Python, SQL.

Build, CI/CD & Tooling: Maven, Gradle, Git, Jenkins, JUnit, Docker, Kubernetes, Airflow/Oozie.

Databases: Oracle, MySQL, HBase.

PROFESSIONAL EXPERIENCE

Onehouse, Bangalore

Open Source Engineer | Sep 2024 – Present

Contributing upstream to Apache Hudi and Apache XTable, designing and optimizing core components for large-scale data lakehouse systems while improving performance, scalability, and reliability across distributed environments.

Key Contributions:

- **AI Spark Diagnostic Agent:** Engineered an interactive Spark Diagnostic Agent that parses application logs, generates automated execution health reports, and fields natural-language queries about job performance enabling rapid bottleneck identification and driving compute cost savings through automated job optimization.
- **Core OSS Contributions:** Authored and reviewed PRs, JIRAs, and RFCs for Apache Hudi and Apache XTable internals, enhancing indexing strategies, compaction, clustering, metadata table performance, and multi-writer concurrency.
- **Customer Cost Optimization:** Engineered an automated metadata and query-log parsing tool that surfaced table-level inefficiencies, enabling enterprise customers to reduce production cloud storage and compute costs by up to 40%.
- **Testing & Benchmarking Architecture:** Designed automated performance and correctness validation suites comparing Hudi execution behavior across Spark, Presto, and Trino engines to gate core releases.
- **Technical Thought Leadership:** Authored deep-dive technical blogs on Hudi internals and lakehouse optimization strategies for the open-source community.

Tech Stack: Java, Scala, Python, Spark, Trino, Presto, AWS (EMR, S3, Athena, Glue), GCP (Dataproc, GCS, Bigquery)

Cloudera, Bangalore

Staff Software Engineer (Spark & Iceberg Backline) | Mar 2020 – Aug 2024

Served as the final escalation point for Cloudera's global support organization—resolved the most complex Spark and Iceberg production outages that frontline support and field engineers could not.

Key Contributions:

- **Spark Production Escalations:** Led root-cause analysis on hundreds of P0 and P1 Spark issues across Cloudera CDP and CDH deployments, diagnosing shuffle failures, OOM patterns, dynamic allocation regressions, Catalyst optimizer edge cases, and Spark-on-YARN integration breaks
- **Iceberg Production Support:** Diagnosed and resolved critical production outages related to **Apache Iceberg** metadata corruption, snapshot isolation conflicts, time-travel query regressions, and partition/schema evolution compatibility across Spark 3.x and Hive Metastore. Routinely debugged Iceberg's integration with Trino and Presto for enterprise customers.
- **Internal Tooling:** Built a unified log-parsing tool that ingests Spark event logs, application logs, **and Iceberg metadata files** to surface stage-level bottlenecks and table-level inefficiencies; adopted by the global support team, drastically reducing average issue resolution time.

- **Cross-Platform Debugging:** Resolved complex integration issues between Spark, Iceberg, and cloud storage (AWS S3, GCS), including consistency model challenges and file listing performance bottlenecks
- **Knowledge Sharing and Training:** Authored and published Spark and Iceberg knowledge articles on the Cloudera Community. Designed and ran Spark Hogwarts, a hands-on internal training program upskilling the frontline support team on Spark internals and modern lakehouse formats (Iceberg and Hudi), reducing escalation rates significantly.

Tech Stack: Java, Scala, Python, Spark, Kafka, Iceberg, Hive, Cloudera CDP, CDH, HDP.

Dell Technologies, Bangalore

Big Data Engineer | Nov 2015 – Mar 2020

Delivered a solution to store data from Support Assist, enabling reporting, support, and information exchange based on customer environments—facilitating a proactive and predictive support environment.

Role and Contributions:

- **Real-Time Telemetry Pipeline:** Designed Spark → Kafka → Storm → HBase ingest path handling unstructured and semi-structured telemetry at scale for the SupportAssist Intelligence Engine.
- **Data Processing:** Read data from Oracle, processed it through Spark, and stored it in Kafka; read data from Kafka queues using Storm for real-time analysis.
- **Data Integration & Analytics:** Designed ingestion and transformation modules using Spark SQL to process complex JSON configurations and publish structured insights to OMNEO UI.
- **Cluster Infrastructure Upgrade:** Co-led the zero-downtime production upgrade of a 40-node Hortonworks HDP cluster (HDP 2.3 to 2.5).
- **Storage Optimization:** Stored large volumes of processed data into HBase, ensuring efficient storage and retrieval for further analysis.

Tech Stack: Hadoop, Hortonworks (HDP 2.6.5), Spark, Storm, Kafka, Oracle, HBase, JSON

Mindtree Pvt. Ltd, Bangalore

Senior Software Engineer | Aug 2014 – Nov 2015

Project: Meeting Services Application - MSA

Built a mobile-and-tablet platform enabling meeting planners to request services without leaving the room.

Role and Contributions:

- **Meeting Services Application:** Built a mobile and tablet platform enabling meeting planners to request services without leaving the room.
- **Backend Development:** Implemented client change requests and developed Controller, Service, and DAO layer components establishing core application logic.
- **Integration and ORM:** Developed Spring Service interfaces and utilized Hibernate for streamlined database operations and Log4j for logging.

Tech Stack: Java, Spring, Hibernate, AngularJS, JQuery, HTML, MySQL

ITApp Software Pvt. Ltd, Bangalore

Junior Software Engineer | Oct 2012 – Jul 2014

Project: AppCenter Cloud Management Platform

A unified cloud services delivery and business management platform enabling cloud services automation for provisioning, lifecycle, monitoring, and user management.

Role and Contributions:

- **AppCenter Cloud Management Platform:** Developed components for a unified cloud services delivery platform enabling automation for provisioning, lifecycle, monitoring, and user management.
- **Component Engineering:** Utilized Spring framework for dependency injection, generated Controller classes for request mapping, created Spring Portlet MVC components, and conducted unit testing with JUnit.

Tech Stack: Java, Spring, RDB, Hibernate, MySQL, Kendo UI, JQuery, HTML, CSS

OPEN SOURCE CONTRIBUTIONS & ACHIEVEMENTS

- **Apache Hudi & XTable Contributor:** Active upstream contributor across indexing, compaction, clustering, and interoperability layers.
- **Cloudera Community Spotlight:** Recognized as a "Rising Star" member with 72 badges and 10+ accepted solutions.
- **Customer Tooling:** Built end-to-end cost-saving tool for Hudi customers (up to 40% cost reduction).
- **Awards:** Recipient of the **Dell Silver Award**, Dell Cash Award for SAIE Analytics, and **Mindtree A-Team Performer Award**.

EDUCATION

- **Master of Computer Applications (MCA)** – Sri Krishnadevaraya University, Anantapur | 2008-2011 | Score: 80%.
- **Bachelor of Science (B.Sc)** – Sri Venkateswara University, Tirupati | 2005-2008 | Score: 81%.